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Importance and Implications of Theory of Bloom's Taxonomy in Different Fields of Education

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Abstract. The theory of Bloom's taxonomy has offered a set of three hierarchical models for cognitive, affective, and psychomotor domains that are used for the classification of educational learning objectives into levels of complexity and specificity. Its cognitive domain as a classification system has been developed to categorize intellectual skills and behavior important to learning. Through its six stages, namely; remembering, understanding, applying, analyzing, evaluating, and creating, this theory emerges different applications from a lower degree to a higher degree. Scholars have examined the taxonomy's main premise, which is that teachers want their students to better understand what they want them to know in an order from simple to sophisticated. The stages are supposed to be hierarchical, with mastery of one level necessitating progression to the next. Many researchers have identified learning objectives that involve higher levels of cognitive capabilities employing Bloom's taxonomy, which focuses on the application of knowledge and abilities to a greater variety of activities and contexts. The data collection technique employed is qualitative method to investigate, and to draw conclusions based on analyzing and synthesizing sources of books and articles on Bloom's Taxonomy. This study examines contribution of Bloom's Taxonomy to evaluation, the teaching-learning process, classroom management, curriculum development, and its various applications in various fields of education. Improving the teaching-learning process and the assessment system at the level of secondary and higher education will give an idea of the application of this taxonomy in different educational aspects.

Keywords: Bloom's taxonomy · Importance · Applications · Education

1 Introduction

Benjamin Bloom and his colleagues released The Taxonomy of Educational Objectives in 1956 as a tool for analyzing learning goals. Bloom's Taxonomy is very well supervised learning model used by elementary and secondary school instructors, university and college learners, and scholars for many years. It provides a unified framework as well as a standard lexicon that educators and researchers can employ. Although some

have incorporated other categories, these taxonomies mainly concentrate on analytical thinking that belongs within the cognitive domain. It can be used in a variety of ways [1]. It's used to determine whether stages of thinking skills are covered by existing instructional content such as learning objectives, curriculum plans, lessons, and evaluations. It can also be used as a guideline for creating educational materials to guarantee that the appropriate cognitive skills are met. It can also be used as a sort of evaluation. This taxonomy can be used in a wide range of educational settings, with the taxonomy being tailored to the particular subject of inquiry. Faculty members should have little issue creating instructional outcomes, customizing technology, and planning classrooms for each course provided they have a basic understanding of Bloom's taxonomy and how to implement it to higher education. It can also be used by educators to break through the clutter of attention by assisting students in setting goals for themselves. This study examines its importance and application in different fields of learning.

2 The Learning Domains of the Bloom's Taxonomy

Bloom's Taxonomy divides academic learning into three categories: cognitive, affective, and psychomotor. The cognitive domain contains mental abilities for knowledge production; the affective domain involves continuous emotional development of mindset; and the psychomotor domain involves physical skills. In 2001, a subsequent edition of the taxonomy adjusts the original version's vocabulary and order of cognitive processes in light of domain findings. According to this reorganization, the ability to synthesize, rather than evaluate, has climbed to the forefront of the hierarchy [2]. Additionally, this alteration imparts a new perspective on all six cognitive abilities. In the diagram [3], the higher levels, the more sophisticated mental processes are allegedly necessary. Higher levels are not always preferable to lower levels, because higher levels cannot be attained without the capacity to use lower levels.

3 Importance and Applications in Different Fields of Education of This Theory

No theory is absolute. There are some limitations in Bloom's theory of taxonomy but the taxonomy of cognitive domain is applied in various segments of learning throughout the decades. One of the most valuable tools accessible to instructors is Bloom's taxonomy. It provides learners with the understanding needed to construct effective evaluation procedures by delivering the answer key to how students learn. It came to a multi-layered answer to the question, which the writers added some explanations to as a teaching tool.

- In a pedagogical exchange, it is important to set learning goals so that both teachers and students know what the exchange is all about.
- The use of templates to define objectives can be beneficial to teachers.
- Planning objectives assist in the clarification of goals for both teachers and pupils.

- Teachers can: “plan and provide suitable lessons”; “create legitimate assessment tasks and techniques”; and “confirm that lesson and evaluation are linked to the objectives” with the assistance of a well-organized set of objectives [4].

Bloom’s taxonomy of learning objectives is used to determine the extent to which an ability or performance has been learnt or mastered. The domain applications linked with each level are mentioned below.

- A learner can define things at their knowledge level of learning.
- A learner can work on specific issues and show off what they’ve learned at the level where they can understand things.
- At the application level of learning, a learner identifies which approaches to employ and then applies those techniques to issues.
- A learner can describe why the solving method works at the analysis level of learning.
- At the synthesis level of learning, a learner can integrate parts of a procedure in new and valuable ways.
- At the evaluation level of learning, a learner can come up with a lot of different ways to deal with the problem.

Then, based on established criteria, the learner can choose the best way to deal with the problem. Bloom’s taxonomy is most typically used to define cognitive learning skills rather than psychomotor or affective capabilities, two categories that are crucial for the achievement of health professionals. It can be used by research scholars who teach or guide others to set learning objectives that define the talents and capabilities they expect their pupils to develop and demonstrate. Bloom’s taxonomy encourages the formulation of learning objectives that concentrate on the abilities that a student must acquire, which has implications for teaching and learning. According to Adams, this system allows for the integration of learning goals that require higher levels of cognitive skills into the system [5]. According to Yildirim and Baur, enhanced class structure, the use of a variety of techniques to lead all learners, increased awareness of learning philosophies, and better supervision of the outputs of instructional strategies are all part of a framework for constructing learning objectives. Problem-based coursework, team projects, and case study are examples of assessment methods that will enhance the learning process [6]. Moreover, according to Dubey, the domains in the original Bloom’s taxonomy can be used for educational reasons, such as developing quality question papers to confirm that students’ cognitive levels are categorized, quality control of a question paper to check up on an evaluation, and question designation for the advancement of automatic question-answering systems [7]. Using the six stages of Bloom’s original taxonomy, the learning procedures can be structured to meet the instructional objectives. Nkhoma et al. discovered that using the amended Bloom’s taxonomy in combination with case-based educational process boosted higher-order thinking. A higher practical assessment could result in more understanding, which could stimulate and motivate study [8]. Bloom’s taxonomy has acquired broad recognition within the assessment community due to its use at United Nations Educational, Scientific, and Cultural Organization (UNESCO) and Organization for Economic Cooperation and Development (OECD) sessions early after

its publication. It's also been used as a basis for curriculum review, exam development, and data analysis.

Teachers and students must understand the cognitive domain's hierarchy of processes and capabilities in order to identify learning skills and how these skills must be adjusted to understand more complex portions of the discipline-specific concept inventory. The enhancement of effective learning should never be assumed when teaching or studying new knowledge. Foundation courses should introduce learners to lower-level process skills, which should then be further improved in intermediate-level courses. Skills associated with higher-level techniques should be effectively implemented and retained in upper-division courses. Another technique to add depth to course exercises while also improving lifetime learning skills is to utilize effective learning skills from different process domains and clusters throughout the cognitive domain. Like the social domain, this aspect serves to remind individuals that their cognitive domain has advanced.

Faculty can be much more particular and concentrate on the cognitive talents students must display in a certain course activity and on an evaluation using Bloom's taxonomy. It also helps teachers figure out how other parts of their course, like subjects, lectures, assignments, and classroom activities, can be better to help students move from lower to higher levels of learning [9]. Bloom's taxonomy supports not only the creation of highly explicit learning objectives and teaching materials in terms of language and definition, but also the assessment of such objectives. In the approach to programming, the process of organizing, educating, learning, and evaluating, there are various inquiries that look into the application of Bloom's taxonomy to teaching and learning. This taxonomy is extremely useful for teachers in correctly implementing the curriculum and assessing students. For example:

- Assessing a unit's or syllabus' objectives gives a clear, simple, visual depiction of the unit or syllabus that may be used to analyze the relative importance assigned to each goal. The Taxonomy's incorporation of labeled, authentic instructional scenarios is one of its strongest features. It gathered accounts of ordinary teaching from teachers to help understand the Taxonomy's divisions and categories, as well as how it may be used to examine objectives.
- Assisting teachers in distinguishing between activities and objectives. The Taxonomy can be used to categorize the learning and teaching activities that were utilized to attain the objectives, as well as the evaluations that were used to indicate how effectively the students learned the objectives [10]. A noteworthy finding emerged from a review of the identified authentic teaching scenarios. Teachers listed actions rather than objectives when citing their unit's outcomes.
- Supporting instructors in understanding the link between evaluation and teaching activities. Airasian and Miranda present an excellent illustration of how teaching affects evaluation [11].

- Examining Curriculum orientation is the procedure of structuring three important parts in a classroom so that they are closely aligned or linked. These are the three aspects: (a) teaching and materials, (b) goals or objectives, and (c) assessments. The idea is that when classroom time and materials are aligned with objectives or guidelines, and these are aligned with examinations, students will learn more efficiently and effectively [12]. Curriculum arrangement is a procedure that enhances the coherence between the written, taught, and assessed curriculum. According to Anderson, the Taxonomy table can be a helpful method for determining curriculum concordance in all subject areas at almost any class or school level. The Taxonomy table can be used with any particular subject by combining themes with categories of knowledge [13].
- Implications for educators and the taxonomy serves to give a framework within which aspiring teachers and instructors can reflect not only how they teach but also how they evaluate and critique their own teaching. They should understand that the only way to assess the effectiveness of their lesson is to look at what students actually understand. As a result, the taxonomy steers aspiring instructors away from "best practice" teaching methods [14].

Within a competence-based curriculum, the Taxonomy functions well as a shareable content in syllabus discussions to reflect various sorts of cognitive capacity. Such arguments address both the curriculum and the students' comprehension of it [15]. In these kinds of classrooms, teachers can use traditional methods to measure how well students have learned, or they can use student self-assessments, or even both [16]. Bloom's taxonomy is especially fascinating in the context of student self-assessment because it relates to double-loop learning, [17] student self-responsibility and self-management, and the student-centered classroom [18]. Bloom's taxonomy has also been used to look at things like effective learning, appropriate faculty interaction, and the long-term design and implementation of global experience-based learning in college teaching. McKeachie based his claim that students learn more when they talk than when they listen to lectures on his own studies [19]. This assertion inspired them to investigate how Bloom's notions may be applied to establish a student-centered educational environment. Self-assessment and increased student participation, which courses are meant to encourage, are important for understanding (Table 1).

Bloom's Taxonomy can be used to determine a student's level of performance in a class. A need to foster further improvement of critical thinking abilities, to ensure understanding with students about their levels of success in classes, and to increase student accountability in courses led us to the taxonomy, moving toward student self-management. The taxonomy is useful for assisting students in taking greater responsibility for their learning and better understanding their critical thinking habits. Scaffolding is teaching that offers support to help the student learn for themselves, and it is used in this way to support metacognition. According to Hogan and Pressley, scaffolding is an instructional component that enables individual pupils with intellectual assistance so that they can operate at the forefront of their cognitive growth. In order to use Bloom's taxonomy as a scaffolding tool, the student must first define the level of his or her effort.

Table 1. Practical tool to use of Bloom's taxonomy for successful teaching.

Levels	Taxonomy	Outcomes	Words (Action Verbs)	Questions
Level 1	Remembering	Students recalling key information	Arrange, Define, Describe, List, Match, Name, Order, Recall, Reproduce	(What, Who, Which) What can you remember about the event?
Level 2	Understanding	Students demonstrate knowledge of the issues acquired while doing so	Classify, Discuss, Explain, Identify, Report, Summarize	(Are you able to write in your own words...? Can you make a quick overview...? What are your thoughts on this....?) Summarize the event in your own words
Level 3	Applying	Focused with pupils' ability to implement their skills and information to a variety of circumstances	Apply, Calculate, Demonstrate, Interpret, Show, Solve, Suggest	(Do you have any further examples of...? Is it possible that this happened in...? Can you adopt the approach to a personal experience...?) Suggest how the main lessons in this event could help other young people
Level 4	Analysing	Pupils' ability to make links between ideas and to analyze critically	Analyse, Appraise, Compare, Contrast, Distinguish, Explore, Infer, Investigate	(What could have happened if...? How did this compare to...? What other results do you perceive as being possible?) Why did the different characters in the event behave the way that they did?

(continued)

Students can then utilize the taxonomy to promote their own higher-level thinking as a result of this self-analysis. Many comprehensive courses, particularly those with critical thinking as a purpose, would seem to benefit from such a method [20].

Table 1. (continued)

Levels	Taxonomy	Outcomes	Words (Action Verbs)	Questions
Level 5	Evaluating	Students can use their experience and wisdom to draw inferences, identify creative solutions to issues, and defend findings	Argue, Assess, Critique, Defend, Evaluate, Judge, Justify	(Is there a smarter method to...? Assess the worth of...? Are you able to defend your viewpoint on...? How efficient are...?) Evaluate the strength of the main character's decision to leave
Level 6	Creating	Students show what they've learned by making something new, whether it's physical or abstract	Compose, Construct, Create, Devise, Generate, Organise, Plan, Produce	(Do you think you can come up with a solution to...? How would you manage with... if you had unlimited resources? Can you think of unique and unexpected ways to use...?) Rewrite the ending of this event, to show a different outcome

Bloom's Taxonomy, when used in higher education, provides a consistent approach to teaching methods while also boosting students' acquisition of subject-specific knowledge. In this exercise, Bloom's Taxonomy is combined with adult education techniques, which enhances students' learning and involvement in the classroom. This tool has been used in a number of college curricula. Positive outcomes have been achieved in each case. The assignment appears to guide students' studying, organizing, and comprehension of course contents, based on their qualitative responses [21].

The use of Bloom's taxonomy in the area of business information systems is well-known around the world [22]. Students' critical thinking abilities develop, and their understanding and application of theory to work improve as a result of the discussion sessions. Likewise, Ching and da Silva found that applying Bloom's taxonomy to a business undergraduate program in a higher education institution was beneficial to both students and teachers [23]. Moreover, Britto and Usman demonstrated the utility of Bloom's taxonomy through an investigation of several studies relevant to software engineering [24]. Bloom's taxonomy was used in the course development and assessment, with four papers publicly acknowledging Bloom's taxonomy's value in software engineering education. Studies on the application of Bloom's taxonomy in the subject of Electrical Engineering at a University of Technology in South Africa, comparable to the

one in this assessment, were conducted during the preparation of curriculum and learner aids in the domain of Electrical Engineering. Academics should attend seminars analyzing learning goals on a regular basis, according to Meda and Swart, 'to verify that the verbs employed in Bloom's taxonomy are used effectively, resulting in increased student learning' [25]. Swart and Daneti looked at the learning results of an electrical fundamentals curriculum that was presented in Romania and South Africa [26]. The usage of Bloom's taxonomy model revealed that the learning results were dominated by the two bottom levels of the taxonomy. The remaining levels were found in the majority of the research, with the authors recommending that other colleges alter learner outcomes to align with Bloom's taxonomy to encourage greater cognitive development.

Bloom's taxonomy is applied to a variety of courses in computer science education. When Michael V. Doran and David D. Langan talk about how to teach Computing I and II, they talk about a method that is based on how people think. This method includes clearly defining and planning course micro-objectives, assigning each micro-objective to a Bloom knowledge level, and helping students achieve and measure their goals [27]. According to Philip Machanick, Bloom's taxonomy is used to build three separate computer science courses: Data and Data Structures, Algorithms and Artificial Intelligence, and Computer Architecture [28]. The Bloom taxonomy was tested by Johnson and Fuller to see if it was suited for computing [29]. Essi Lahtinen provides the findings of research on 254 undergraduate students in a basic programming course whose achievement was assessed on various levels of Bloom's Taxonomy, as well as the outcomes of numerical cluster analysis, which recommends that the groups of students acquired align with Bloom's taxonomy: underperforming students at lower levels may still perform well at higher levels of taxonomy. As a result, Lahtinen proposes that while structuring basic programming courses, the instructor should distinguish six kinds of pupils: capable students, practical students, poorly prepared students, theoretical students, memorizing students, and indifferent students. Theoretical students, for example, did well at the highest level (evaluation), but had a hard time with application and synthesis [30].

In terms of Bloom's Taxonomy, Gardner and colleagues discovered the following Merge and Quick Sorts Learning Goals: (1) The learner will understand how to perform a fusion process. (2) The student will understand how to use the operations in order. (3) The student will understand the fusion process and quick sort Big O analysis. (4) The student will assess each of Kruse's five sorting strategies [31]. Andrew Churches proposed a "digital" taxonomy for 21st-century schools and students. He rewrote the main verbs that indicate cognitive aims, combining creative and classic words. Because digital technology has become a vital component of education and other aspects of life, A. Churches' modification allows Bloom's pedagogical taxonomy to be extended by bringing it nearer to the present realities of human life [32]. Diana Cukierman and Donna McGee Thompson present the results of their research into the impact of integrating learning strategies teaching into standard laboratory time courses to aid student learning. Learning techniques and students were exposed to Bloom's learning stages taxonomy and taught how this approach applies to course content in the first session.

4 Conclusion

Finally, Bloom's taxonomy for educational objectives can be regarded as having made a significant contribution to teaching and learning. Bloom's Taxonomy is a useful framework for instructors to employ in order to concentrate on higher-order thinking and gives a hierarchy of levels. This taxonomy can help teachers create performance assignments, prepare questions for student consultations, and provide assessment. Educational psychologist Benjamin Bloom classified what and how humans learn into three distinct domains of learning in the mid-twentieth century. Content knowledge and the enhancement of human talents are included in the Cognitive Domain. Using the taxonomy at all levels of education, from kindergarten to college, and across different fields shows that it can be used in a wide range of different fields [33]. When critical thinking is used in higher levels of learning, the process of learning is both sequential and fluid. In other words, rather than being purely hierarchical, the cognitive action of assessing new knowledge, synthesizing valuable learning with other information, and then assessing the pieces of that and acquiring new knowledge is continual and interconnected. Bloom's taxonomy is widely used in the field of education. In order to better comprehend this learning, each field must use and apply it. Bloom's taxonomy has been used in a variety of fields, including biology, [34] management, [35] and music [36]. Bloom's taxonomy is widely regarded as a well-defined strategy for instructors to use since it improves students' understanding of and capacity to analyze, perform, and understand. It's a way to organize educational goals in a way that makes it easier to improve educational instruction and come up with ways to measure educational progress. It has been used to learn in the fields of science, engineering, business, humanities, and social science. Many scholars use the technique to explicitly establish learning objectives, while others use it to evaluate students' understanding and create learning materials and evaluations. Bloom's taxonomy is fundamental knowledge for all educational and cognitive psychology programs. Though it was originally created as a simple evaluation tool, it has evolved into a symbol for curriculum planning, being used to define learning objectives and plan activities in the classroom. It has been developed for use in primary and secondary school classes, as well as in every academic area.

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